

Characterizing Random Signals

From individual waveforms to ensembles, stationarity, and correlation

$$\mu_x = E\{x[n]\}$$

$$r_{xx}[m] = E\{x[n]x[n+m]\}$$

Roadmap and learning objectives

75 minutes • stochastic-process vocabulary → WSS → correlation → ML alignment

- Distinguish a random variable, a random process, an ensemble, and one realization.
- Explain strict-sense stationarity (SSS) and wide-sense stationarity (WSS).
- Compute / interpret mean, autocorrelation, cross-correlation, and correlation matrices.
- Use correlation as a feature-alignment and template-matching operation before ML.

0–15 min

Stochastic processes and ensembles

15–35 min

SSS vs WSS; what “stationary” buys us

35–60 min

Mean, correlation, matrices

60–75 min

ML bridge: alignment

01

From a random variable to a random process

A signal value is random; a whole sequence of random variables is a stochastic process.

Random process = indexed family of random variables

At each time index n , $X[n]$ has a distribution; a realization picks one value at every n .

$X[n]$: random variable at each n

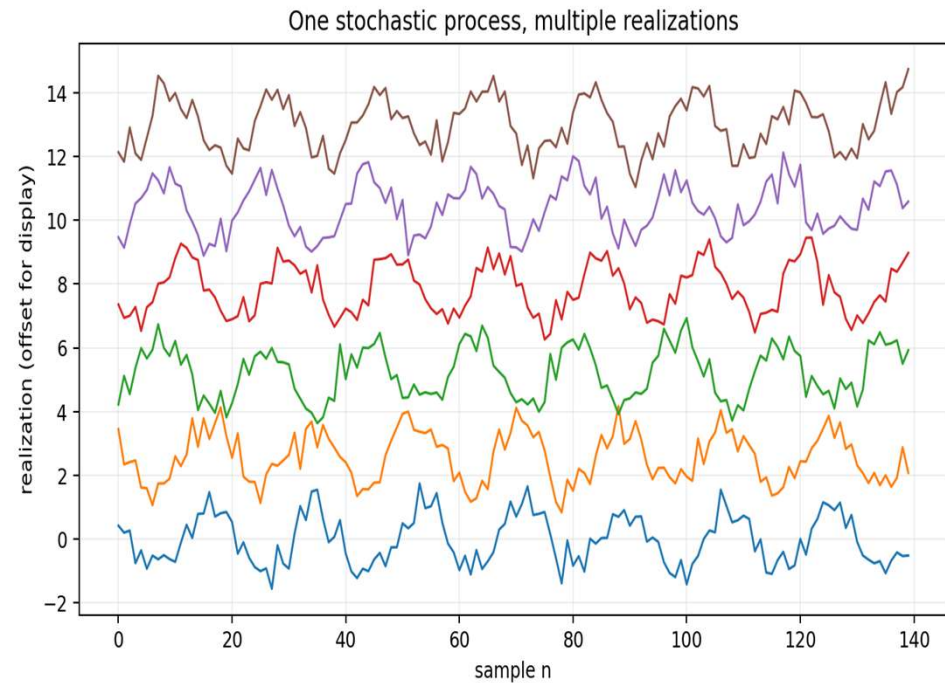
$x_i[n]$: the i -th realization (sample path)

Ensemble

The set of all possible realizations generated by the same probabilistic mechanism.

One realization

What a sensor normally gives us: one finite sample path, not the whole ensemble.



Ensemble average vs time average

Expectation is defined over the ensemble; experiments often estimate it from time samples.

$$\mu_x[n] = E\{X[n]\}$$

$$\text{time estimate: } \hat{\mu} = (1/N) \sum_n x[n]$$

- These are not automatically the same thing.
- Stationarity removes explicit dependence on n for key statistics.
- Ergodicity is the additional idea that lets time averages estimate ensemble averages.

Teaching nuance

Do not say “WSS means time averages equal expectations.” That is ergodicity, not stationarity.

02

Stationarity

Can the process statistics be treated as unchanged when we shift the time origin?

Strict-sense stationarity (SSS)

Every joint probability distribution is invariant to a common time shift.

$$(X[n_1], \dots, X[n_k]) \approx (X[n_1 + \ell], \dots, X[n_k + \ell])$$

- The strongest common stationarity definition.
- It constrains all orders of statistics, not just the mean and variance.
- Powerful mathematically, but often more than a DSP design actually needs.

Example

Independent identically distributed Gaussian noise with fixed mean and variance is SSS.

Non-example

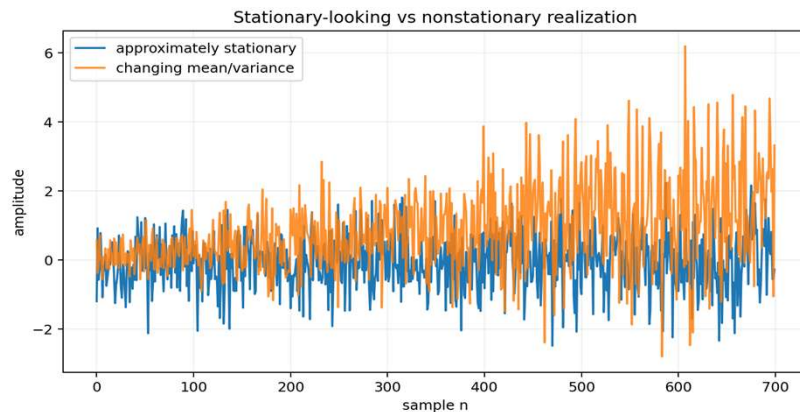
A process whose variance grows with n is not stationary.

Wide-sense stationarity (WSS)

Most linear DSP analysis only needs first- and second-order statistics.

$$E\{X[n]\} = \mu_x \text{ (constant)}$$

$$r_{xx}[n_1, n_2] = r_{xx}[n_2 - n_1]$$



- Finite second moment.
- Mean does not depend on absolute time.
- Autocorrelation depends only on lag, not on where the pair occurs.
- Exactly the structure needed for correlation, power, and LTI-system analysis.

Why ML cares

Stable preprocessing statistics and lag features are much easier to learn from when the data distribution is approximately stationary.

CONCEPT CHECK

A process has constant mean but variance $\sigma^2[n] = 1 + 0.01n$. Is it WSS?

Good answer

No. WSS requires the second-order statistics to be time-shift invariant. A variance that changes with n violates that requirement even though the mean is constant.

03

Time-domain statistics

Mean describes level; correlation describes similarity across lag.

Mean and variance: what fluctuates around what?

For WSS, these do not depend on n .

$$\mu_x = E\{X[n]\}$$

$$\sigma_x^2 = E\{|X[n] - \mu_x|^2\}$$

$$r_{xx}[0] = E\{|X[n]|^2\} = \sigma_x^2 + |\mu_x|^2$$

Interpretation

Mean = DC / baseline level. Variance = average squared fluctuation around that level.

Zero-mean case

If $\mu_x = 0$, then $r_{xx}[0] = \sigma_x^2$. This is why white-noise autocorrelation is often written $\sigma^2 \delta[m]$.

Finite data

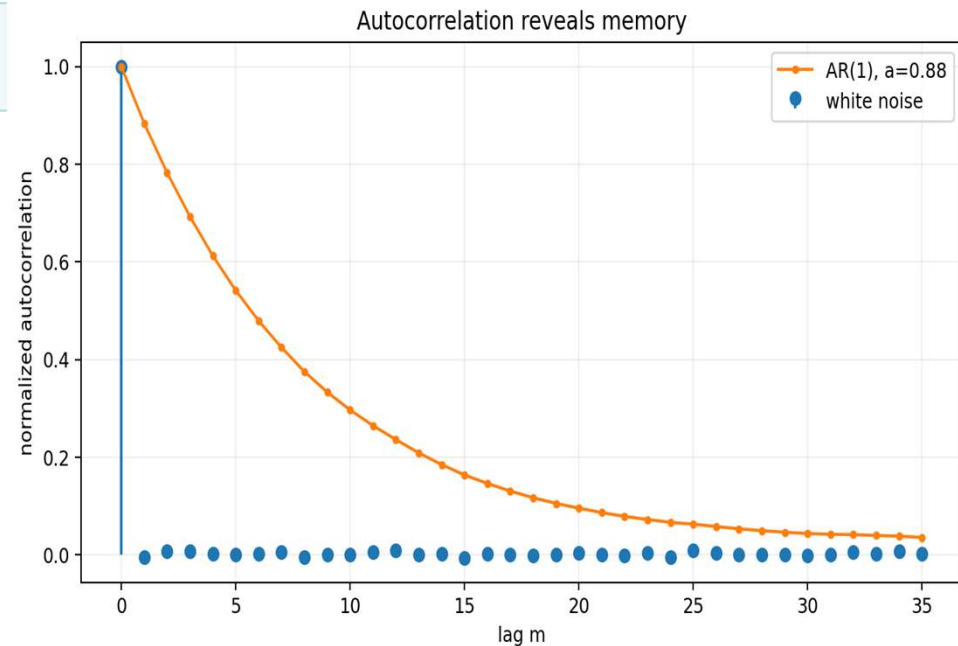
Use sample estimates, but remember they are random quantities with bias/variance of their own.

Autocorrelation: how similar is the process to a delayed copy?

With the convention used here: $r_{xx}[m] = E\{X[n]X^*[n+m]\}$.

$$r_{xx}[m] = E\{X[n] X^*[n+m]\}$$

- Large $|r_{xx}[m]|$ means samples m apart are statistically related.
- White noise: correlation collapses to lag 0.
- Smooth / filtered processes keep correlation over several lags.
- For real WSS processes, $r_{xx}[m] = r_{xx}[-m]$.

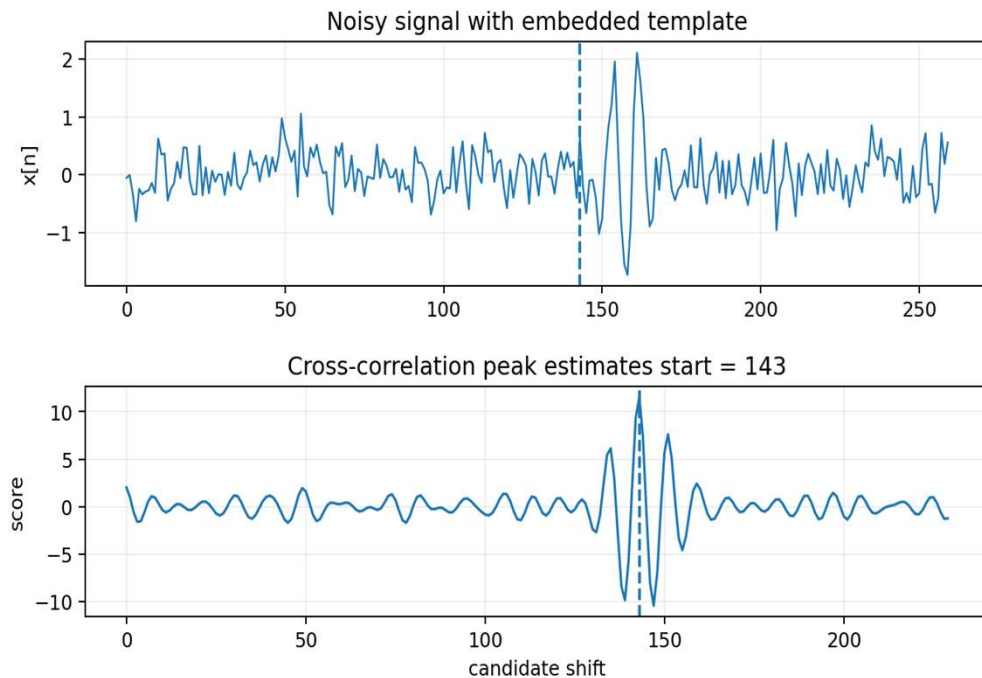


Cross-correlation: similarity between two signals as a function of lag

Our convention: $r_{xy}[m] = E\{X[n]Y^*[n+m]\}$.

$$r_{xy}[m] = E\{X[n] Y^*[n+m]\}$$

- Peak lag estimates relative delay / alignment.
- Useful for synchronization, matched-template detection, and feature registration.
- Sign conventions differ across books/software; define the convention before deriving formulas.

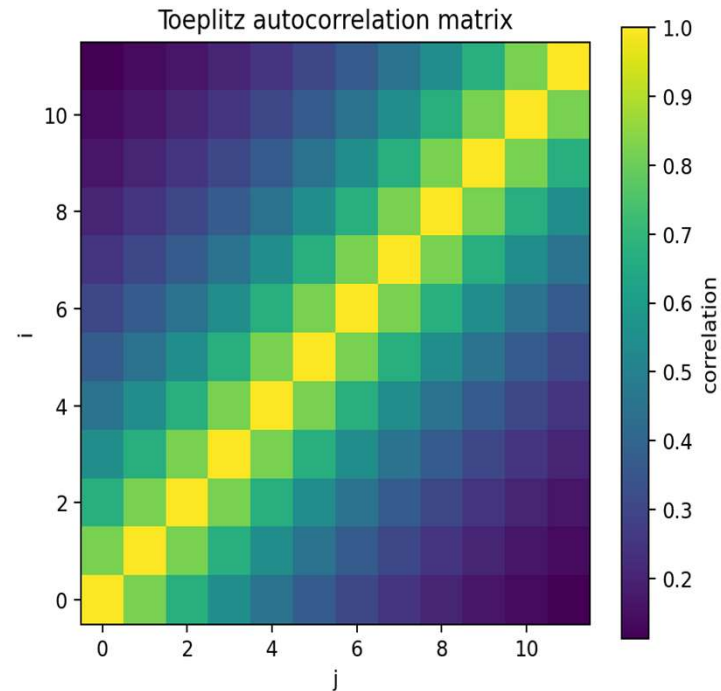


The autocorrelation matrix packages many lags at once

A finite vector of WSS samples has a Toeplitz covariance/correlation structure.

$$R_x[i,j] = r_{xx}[j-i]$$

- Hermitian / symmetric for complex / real processes.
- Toeplitz for WSS: entries depend only on lag.
- Positive semidefinite: $\mathbf{a}^H \mathbf{R}_x \mathbf{a} \geq 0$ for every vector $\mathbf{a}$.
- Eigenvalues cannot be negative (up to numerical roundoff).



CONCEPT CHECK

Why must an autocorrelation matrix be positive semidefinite?

Good answer

For any coefficients a , the quadratic form $a^H R_x a$ equals the expected squared magnitude $E\{|\sum a_k^* X[k]|^2\}$. A squared magnitude cannot have negative expectation.

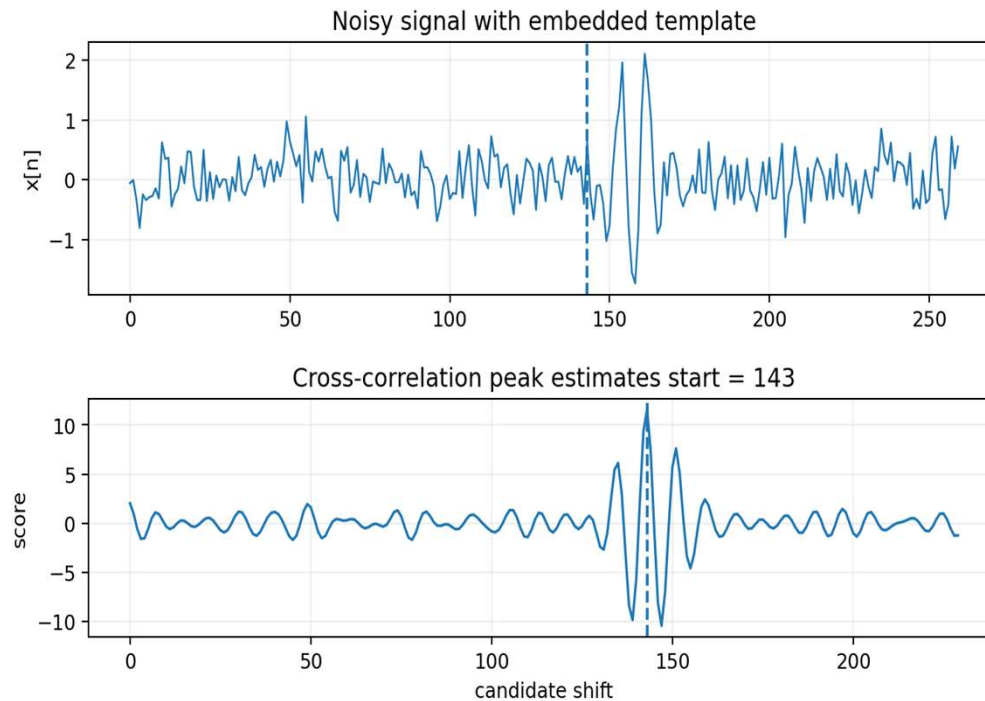
04

ML bridge: alignment before prediction

Correlation can turn “same event at different times” into aligned features.

Template matching as preprocessing

Correlation converts unknown time shift into an estimated alignment parameter.



- Compute correlation between an observation and a known template.
- Find the lag with the largest correlation score.
- Shift / crop the example into a common coordinate system.
- Then compute features and feed fixed-size vectors to a Scikit-Learn model.

Important

Correlation is not a classifier by itself; here it is an alignment / feature-engineering step.

Coding prompt: estimate lag from noisy data

Predict first; then run the notebook.

1

Generate

Insert a known pulse / waveform at an unknown offset inside noise.

2

Correlate

Use `scipy.signal.correlate(observation, template)`.

3

Estimate

`argmax` gives the best candidate alignment under this simple model.

What to discuss

How does SNR change the peak? What if the template amplitude changes? What if the signal is stretched, not merely shifted? Why might normalized correlation be preferable?

Lecture 15 wrap-up

The stochastic framework lets us describe what is repeatable even when individual samples are unpredictable.

- A random process is an ensemble; a sensor usually provides one realization.
- WSS means constant mean and lag-only autocorrelation.
- Autocorrelation measures memory; cross-correlation measures lagged similarity.
- Correlation matrices are symmetric/Hermitian, Toeplitz (for WSS), and positive semidefinite.
- Correlation can align examples before ML feature extraction.

EXIT QUESTION

A process has constant mean and $r_{xx}[n_1, n_2] = \exp(-0.2|n_2 - n_1|)$. What property makes it WSS?

Good answer

Its mean is constant and its autocorrelation depends only on lag $n_2 - n_1$, not absolute time.

References and instructor resources

- Oppenheim & Schaffer, Discrete-Time Signal Processing: random signals, correlation, and LTI systems.
- Raschka et al., Machine Learning with PyTorch and Scikit-Learn: preprocessing and PyTorch workflows.
- Accompanying notebook: `Week8_Random_Signals_DSP_ML_Demos.ipynb`.
- Homework and weekly quiz: `Week8_Random_Signals_Homework_Quiz_Solutions.docx`.